

Lab 18– API Integration: Connecting to external services with error handling

Name: D.ANKITHA

Enrollment Number: 2503A51L09

Assignment Number:18.2

Lab Question 1 : Weather Forecasting API

Task 1

Prompt:

Write a Python script using the OpenWeatherMap API to fetch and display current temperature and weather description for a given city with error handling for missing or invalid API key.

Code:

```
assignment 18.2 > lab q1 > task.py > ...
1  """
2  Simple Python Weather App
3  Fetches and displays the current temperature and weather description
4  for any city using the OpenWeatherMap API.
5  """
6
7  import requests
8
9  # 🗝️ Your personal API key (replace this with your actual key)
10 API_KEY = "ee15900b8a955f7c9c7d2f83f02ae831"
11 API_URL = "https://api.openweathermap.org/data/2.5/weather"
12
13
14 def fetch_weather(city: str) -> dict:
15     """Fetch weather data from OpenWeatherMap API for the given city."""
16     params = {"q": city, "appid": API_KEY, "units": "metric"}
17
18     try:
19         response = requests.get(API_URL, params=params, timeout=10)
20         response.raise_for_status()
21     except requests.exceptions.RequestException as e:
22         print(f"⚠️ Network error: {e}")
23         return None
24
25     data = response.json()
26
27     if data.get("cod") != 200:
28         print(f"❌ Error: {data.get('message', 'Unknown error')}")
29         return None
30
31     return data
32
33
34 def main():
35     print("☁️ Welcome to the Simple Weather App ☁️")
36     city = input("Enter city name: ").strip()
37
38     if not city:
39         print("❗ Please enter a valid city name.")
40         return
41
42     weather_data = fetch_weather(city)
43
44     if weather_data:
45         temp = weather_data["main"]["temp"]
46         description = weather_data["weather"][0]["description"].capitalize()
47         print(f"\n 🌡️ {city}")
48         print(f" 🌤️ Temperature: {temp}°C")
49         print(f" 🌤️ Weather: {description}")
50     else:
51         print("Failed to fetch weather data.")
52
53
54 if __name__ == "__main__":
55     main()
```

Output:

```
2/lab q1/task.py"
🌤️ Welcome to the Simple Weather App 🌤️
Enter city name: warangal

📍 warangal
🌡️ Temperature: 24.61°C
☁️ Weather: Overcast clouds
PS C:\Users\Subana_Behan\OneDrive\Desktop\A
```

Observations:

- I used AI to help me understand how to send API requests using the requests module.
- It also showed me how to check if my API key is missing or wrong.
- I learned that JSON responses can be accessed like dictionaries in Python.

Task 2

Prompt:

Extend the weather script to save weather data into a CSV file without duplicates and handle file read/write errors.

Code:

```
background 182 > labq1 > task2.py > ...
1
2 Simple Weather App 🌤️
3 Fetches and displays the current temperature and weather description for any city
4 using the Openweathermap API, and saves data into a CSV file (without duplicates).
5
6
7
8
9
10
11
12 API_KEY = "a1c10b0a055f79c7d3f83f82a031"
13 API_URL = "https://api.openweathermap.org/data/2.5/weather"
14 CSV_FILE = "weather_data.csv"
15
16
17 def fetch_weather(city: str) -> dict | None:
18     """Fetch weather data from Openweathermap API for the given city."""
19     params = {"q": city, "appid": API_KEY, "units": "metric"}
20
21     try:
22         response = requests.get(API_URL, params=params, timeout=10)
23         data = response.json()
24
25         # Handle invalid city
26         if response.status_code == 404 or data.get("cod") == "404":
27             print(f"❌ City '{city}': not found. Please check the spelling.")
28             return None
29
30     except requests.exceptions.RequestException as e:
31         print(f"⚠️ Network error: {e}")
32         return None
33
34
35 def save_weather_to_csv(city: str, temp: float, desc: str) -> None:
36     """Save weather data to CSV file, avoiding duplicate city entries."""
37     # Ensure file exists, or create it with headers
38     try:
39         file_exists = os.path.exists(CSV_FILE)
40         if not file_exists:
41             # Read existing city names to prevent duplicates
42             if file_exists:
43                 with open(CSV_FILE, mode="r", newline="", encoding="utf-8") as f:
44                     reader = csv.DictReader(f)
45                     cities = {row["city"].lower() for row in reader}
46
47             if city.lower() in cities:
48                 print(f"⚠️ Weather for '{city}' already exists in {CSV_FILE}. Skipping duplicate entry.")
49                 return
50
51             # Append new data
52             with open(CSV_FILE, mode="a", newline="", encoding="utf-8") as f:
53                 fieldnames = ["city", "temperature (°C)", "weather description"]
54                 writer = csv.DictWriter(f, fieldnames=fieldnames)
55
56                 if not file_exists:
57                     writer.writeheader()
58
59                 writer.writerow({
60                     "city": city,
61                     "temperature (°C)": temp,
62                     "weather description": desc
63                 })
64
65         print(f"✅ Saved weather data for '{city}' to {CSV_FILE}!")
66     except (OSError, IOError) as e:
67         print(f"⚠️ File error while saving data: {e}")
68
69
70 def main():
71     print("🌤️ Welcome to the Simple Weather App 🌤️")
72     city = input("Enter city name: ").strip()
73
74     if not city:
75         print("❗ Please enter a valid city name.")
76         return
77
78     weather_data = fetch_weather(city)
79
80     if weather_data:
81         temp = weather_data["main"]["temp"]
82         description = weather_data["weather"][0]["description"].capitalize()
83
84         print(f"📍 {city}")
85         print(f"🌡️ Temperature: {temp}°C")
86         print(f"☁️ Weather: {description}")
87
88         save_weather_to_csv(city, temp, description)
89     else:
90         print(f"❌ Could not fetch weather data.")
91
92
93 if __name__ == "__main__":
94     main()
95
```

Output:

```
Welcome to the Simple Weather App 🌤️
Enter city name: warangal

📍 warangal
🌡️ Temperature: 24.61°C
☁️ Weather: Overcast clouds
💾 Weather for 'warangal' already exists in weather_data.csv. Skipping duplicate entry.
```

```
weather_data.csv > data
1 City, Temperature (°C), Weather Description
2 warangal, 24.61, Overcast clouds
3
```

Observations:

- I learned how to use csv.DictWriter to store data safely.
- The AI helped me avoid duplicate city entries by checking before writing.
- I also handled file errors using try and except blocks.

Lab Question 2: Currency Exchange Rate API

Task 1

Prompt:

Write a Python script that takes user input (amount, source, target currency) and fetches the latest exchange rate with error handling for invalid currency codes.

Code:

```

assignment 18.2 > lab2 > task1.py > fetch_exchange_rate
1 import requests
2
3 API_KEY = "fded72fb50ef5a4d05aaa13c1d95692e"
4 BASE_URL = "https://open.er-api.com/v6/latest/" # reliable free exchange rate API
5
6 def fetch_exchange_rate(source_currency, target_currency):
7     """Fetch exchange rate from source to target currency."""
8     try:
9         response = requests.get(f"{BASE_URL}{source_currency.upper()}")
10        response.raise_for_status() # raise error for bad responses
11        data = response.json()
12
13        if data.get("result") != "success":
14            raise ValueError("Invalid source currency or API error.")
15
16        rates = data.get("rates", {})
17        if target_currency.upper() not in rates:
18            raise ValueError("Invalid target currency code.")
19
20        return rates[target_currency.upper()]
21
22    except requests.exceptions.RequestException as e:
23        print("▲ Network or connection error:", e)
24    except ValueError as e:
25        print("▲ ", e)
26    except Exception as e:
27        print("▲ Unexpected error:", e)
28
29    return None
30
31
32 def main():
33     try:
34         amount = float(input("Enter amount: "))
35         source = input("Enter source currency (e.g., USD, EUR, INR): ").strip()
36         target = input("Enter target currency (e.g., USD, EUR, INR): ").strip()
37
38         rate = fetch_exchange_rate(source, target)
39         if rate:
40             converted = amount * rate
41             print(f"\n💱 {amount:.2f} {source.upper()} = {converted:.2f} {target.upper()}")
42         else:
43             print("Conversion failed. Please check your currency codes.")
44     except ValueError:
45         print("▲ Please enter a valid numeric amount.")
46
47
48 if __name__ == "__main__":
49     main()
50

```

Output:

```

Enter amount: 1000
Enter source currency (e.g., USD, EUR, INR): usd
Enter target currency (e.g., USD, EUR, INR): inr

💱 1000.00 USD = 88250.21 INR

```

Observations:

- I used AI to help create input prompts and handle wrong currency codes.
- I understood how to check if the API response was valid before using it.
- The script now gives a clear error message when something goes wrong.

Task 2

Prompt:

Add retry logic to the currency script to attempt the API call up to three times if it fails, and log all errors into a local file.

Code:

```

assignment 18.2 > lab2 > task2.py > ...
1  import requests
2  import time
3  import logging
4
5  # --- Configuration ---
6  API_KEY = "fdded72fb50ef54dd85aaa13cid95692e"
7  BASE_URL = "https://open.er-api.com/v6/latest/"
8  MAX_RETRIES = 3
9  LOG_FILE = "error_log.txt"
10
11 # --- Logging Setup ---
12 logging.basicConfig(
13     filename=LOG_FILE,
14     level=logging.ERROR,
15     format="%asctime)s - %(levelname)s - %(message)s"
16 )
17
18
19 def fetch_exchange_rate(source_currency, target_currency):
20     """Fetch exchange rate with retry logic and error logging."""
21     for attempt in range(1, MAX_RETRIES + 1):
22         try:
23             print(f"Attempt {attempt} to fetch exchange rate...")
24             response = requests.get(f"{BASE_URL}{source_currency.upper()}", timeout=10)
25             response.raise_for_status()
26
27             data = response.json()
28             if data.get("result") != "success":
29                 raise ValueError("Invalid source currency or API returned an error.")
30
31             rates = data.get("rates", {})
32             if target_currency.upper() not in rates:
33                 raise ValueError("Invalid target currency code.")
34
35             return rates[target_currency.upper()]
36
37         except requests.exceptions.RequestException as e:
38             logging.error(f"Network error (Attempt {attempt}): {e}")
39             print(f"Network issue on attempt {attempt}. Retrying...")
40             time.sleep(2)
41
42         except ValueError as e:
43             logging.error(f"Value error: {e}")
44             print(f"Value error: {e}")
45             break # no need to retry invalid currency codes
46
47         except Exception as e:
48             logging.error(f"Unexpected error: {e}")
49             print(f"Unexpected error occurred.")
50             break
51
52     print(f"Failed to fetch exchange rate after multiple attempts.")
53     return None
54
55
56 def main():
57     try:
58         amount = float(input("Enter amount: "))
59         source = input("Enter source currency (e.g., USD, EUR, INR): ").strip()
60         target = input("Enter target currency (e.g., USD, EUR, INR): ").strip()
61
62         rate = fetch_exchange_rate(source, target)
63         if rate:
64             converted = amount * rate
65             print(f"\nQ (amount:.2f) {source.upper()} = {converted:.2f} {target.upper()}")
66         else:
67             print("Conversion failed. Please check your inputs or try again later.")
68     except ValueError:
69         print(f"Please enter a valid numeric amount.")
70         logging.error("Invalid numeric amount entered by user.")
71
72 if __name__ == "__main__":
73     main()
74
75

```

Output:

```

assignment 18.2/lab2/task2.py
Enter amount: 49999
Enter source currency (e.g., USD, EUR, INR): kin
Enter target currency (e.g., USD, EUR, INR): rur
Attempt 1 to fetch exchange rate...
Invalid source currency or API returned an error.
Failed to fetch exchange rate after multiple attempts.
Conversion failed. Please check your inputs or try again later.
error_log.txt
1 2025-10-28 13:08:20,916 - ERROR - Invalid numeric amount entered by user.
2 2025-10-28 13:08:45,954 - ERROR - Value error: Invalid source currency or API returned an error.
3

```

Observations:

- AI showed me how to use a simple retry loop with for and try.
- I created an error log file using basic file handling.
- I learned that logging helps when APIs fail due to server issues.

Lab Question 3 – News Headlines API

Task 1

Prompt:

Write a Python script to fetch and print the top 5 technology headlines from a news API, with timeout error handling.

Code:

```
assignment 18.2 > lab3 > task1.py > fetch_tech_news
1  import requests
2
3  API_KEY = "9cac00a9af334958987e781c4a196c75"
4  URL = "https://newsapi.org/v2/top-headlines"
5
6  def fetch_tech_news():
7      params = {
8          "category": "technology",
9          "language": "en",
10         "pageSize": 5, # fetch only top 5 headlines
11         "apiKey": API_KEY
12     }
13
14     try:
15         response = requests.get(URL, params=params, timeout=5)
16         response.raise_for_status() # raise error for bad status codes
17         data = response.json()
18
19         if data.get("status") != "ok":
20             print("▲ Error fetching news:", data.get("message", "Unknown error"))
21             return
22
23         print("\n■ Top 5 Technology Headlines:\n")
24         for i, article in enumerate(data.get("articles", []), 1):
25             print(f"{i}. {article['title']}")
26             print(f"    Source: {article['source']['name']}")
27             print(f"    URL: {article['url']}\n")
28
29     except requests.Timeout:
30         print("⌚ Request timed out. Please check your internet connection and try again.")
31     except requests.RequestException as e:
32         print(f"✖ Network error: {e}")
33     except Exception as e:
34         print(f"▲ Unexpected error: {e}")
35
36 if __name__ == "__main__":
37     fetch_tech_news()
38
```

Output:

```
1  C:\Users\lab3\OneDrive\Desktop\lab3\Task1\Task1.py
2  ■ Top 5 Technology Headlines:
3
4  1. Retroid Pocket 6 and Pocket G2 Officially Unveiled - Retro Handhelds
5     Source: Retrohandhelds.gg
6     URL: https://retrohandhelds.gg/retroid-pocket-6-and-pocket-g2-officially-unveiled/
7
8  2. This Week's Japanese Game Releases: Dragon Quest I & II HD-2D Remake, Tales of Xillia Remastered, more - Gematsu
9     Source: Gematsu
10    URL: https://www.gematsu.com/2025/10/this-weeks-japanese-game-releases-dragon-quest-i-ii-hd-2d-remake-tales-of-xillia-remastered-more
11
12  3. Cancelled God of War PS5 Game Leaks in Screenshots - Push Square
13     Source: Push Square
14     URL: https://www.pushsquare.com/news/2025/10/cancelled-god-of-war-ps5-game-leaks-in-screenshots
15
16  4. Xbox take massive 200+ foot TV into the air over Miami to break bizarre gaming world records - supercarblondie.com
17     Source: Supercarblondie.com
18     URL: https://supercarblondie.com/xbox-ninja-gaiden-4-helicopter-record-swae-lee/
19
20  5. The Neuroscience-Based Nike Mind 001 Appears in "Black" - hypebeast.com
21     Source: HYPEBEAST
22     URL: https://hypebeast.com/2025/10/nike-mind-001-black-hq4307-001-release-info
```

Observations:

- AI helped me understand how to use the timeout parameter in requests.
- The script now prints headlines clearly in the console.
- I learned how to handle slow or failed responses without crashing.

Task 2

Prompt:

Clean and preprocess the headlines by removing special characters and converting them to title case, handling empty or null values.

Code:

```
assignment 18.2 > lab3 > task2.py > fetch_tech_news
1  import requests
2  import re
3  API_KEY = "9cac00a9af334958987e781c4a196c75"
4  URL = "https://newsapi.org/v2/top-headlines"
5  def clean_headline(text):
6      """Remove special characters and convert to title case."""
7      if not text or not isinstance(text, str):
8          return "No Title Available"
9      # Remove special characters except letters, numbers, spaces, and basic punctuation
10     cleaned = re.sub(r"[^A-Za-z0-9 .!?'-]", "", text)
11     return cleaned.title()
12 def fetch_tech_news():
13     params = {
14         "category": "technology",
15         "language": "en",
16         "pageSize": 5,
17         "apiKey": API_KEY
18     }
19     try:
20         response = requests.get(URL, params=params, timeout=5)
21         response.raise_for_status()
22         data = response.json()
23         if data.get("status") != "ok":
24             print("⚠ Error fetching news:", data.get("message", "Unknown error"))
25             return
26         print("\n📰 Top 5 Cleaned Technology Headlines:\n")
27         for i, article in enumerate(data.get("articles", []), 1):
28             raw_title = article.get("title", "")
29             clean_title = clean_headline(raw_title)
30             source = article.get("source", {}).get("name", "Unknown Source")
31             url = article.get("url", "No URL Provided")
32             print(f"{i}. {clean_title}")
33             print(f"    Source: {source}")
34             print(f"    URL: {url}\n")
35     except requests.Timeout:
36         print("⏰ Request timed out. Please check your internet connection and try again.")
37     except requests.RequestException as e:
38         print(f"❌ Network error: {e}")
39     except Exception as e:
40         print(f"⚠ Unexpected error: {e}")
41 if __name__ == "__main__":
42     fetch_tech_news()
43
```

Output:

Top 5 Cleaned Technology Headlines:

1. Retroid Pocket 6 And Pocket G2 Officially Unveiled - Retro Handhelds
Source: Retrohandhelds.gg
URL: <https://retrohandhelds.gg/retroid-pocket-6-and-pocket-g2-officially-unveil>
2. This Weeks Japanese Game Releases Dragon Quest I Ii Hd-2D Remake, Tales Of Xill
Source: Gematsu
URL: <https://www.gematsu.com/2025/10/this-weeks-japanese-game-releases-dragon-d>
e
3. Cancelled God Of War Ps5 Game Leaks In Screenshots - Push Square
Source: Push Square
URL: <https://www.pushsquare.com/news/2025/10/cancelled-god-of-war-ps5-game-leak>
4. Xbox Take Massive 200 Foot Tv Into The Air Over Miami To Break Bizarre Gaming W
Source: Supercarblondie.com
URL: <https://supercarblondie.com/xbox-ninja-gaiden-4-helicopter-record-swae-lee>
5. The Neuroscience-Based Nike Mind 001 Appears In Black - Hypebeast.Com
Source: HYPEBEAST
URL: <https://hypebeast.com/2025/10/nike-mind-001-black-hq4307-001-release-info>

Observations:

- I used AI to clean text using regular expressions and title() function.
- It helped me skip empty or None headlines safely.
- The cleaned headlines look much more readable now.

-